

Find basis spanning Hodge spaces

- **Nonzero** Eigenspace of **down Laplacian** spans the **gradient** space
- **Nonzero** Eigenspace of **up Laplacian** spans the **curl** space
- **Kernel** of Laplacian spans the **harmonic** space

Simplicial Fourier transform

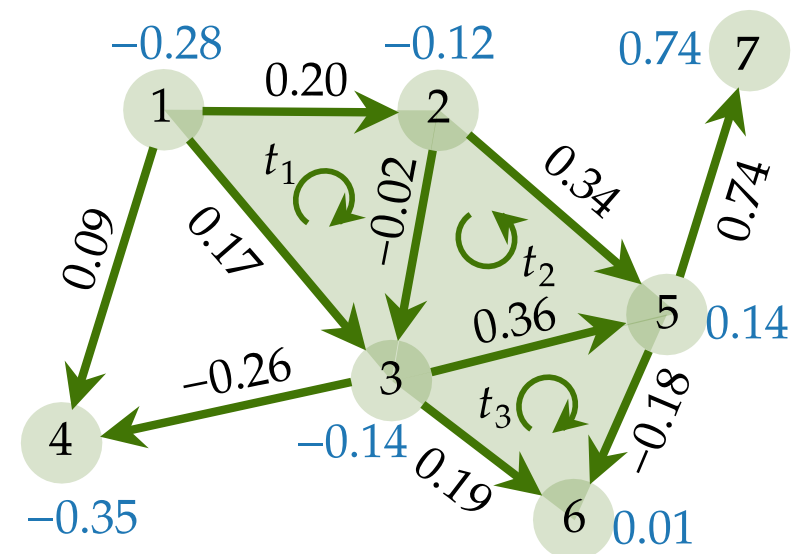
Frequency — eigenvalues

Fourier basis — eigenvectors

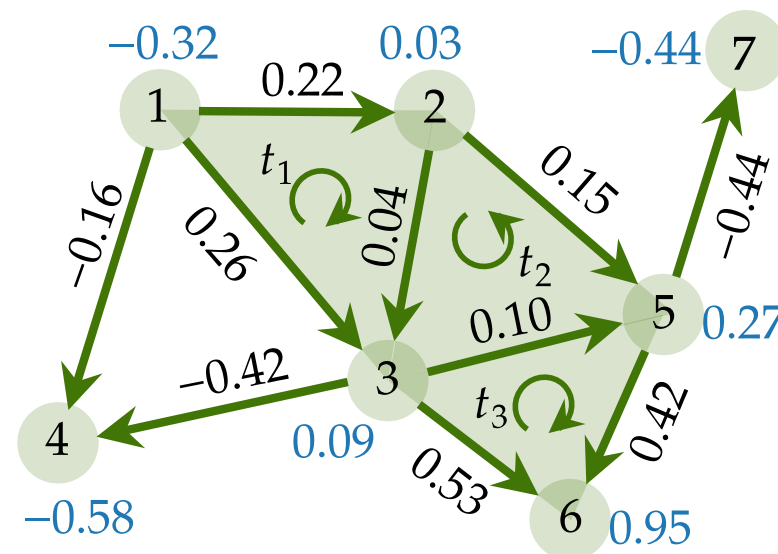
$$\lambda_G = \|\mathbf{B}_1 \mathbf{u}_G\|_2^2$$

Gradient eigenvector

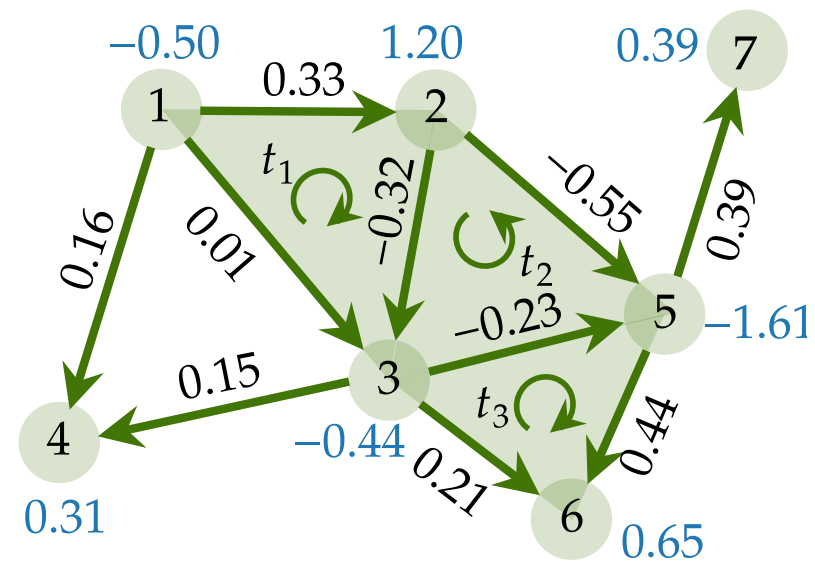
Fourier basis reflecting **divergent** properties



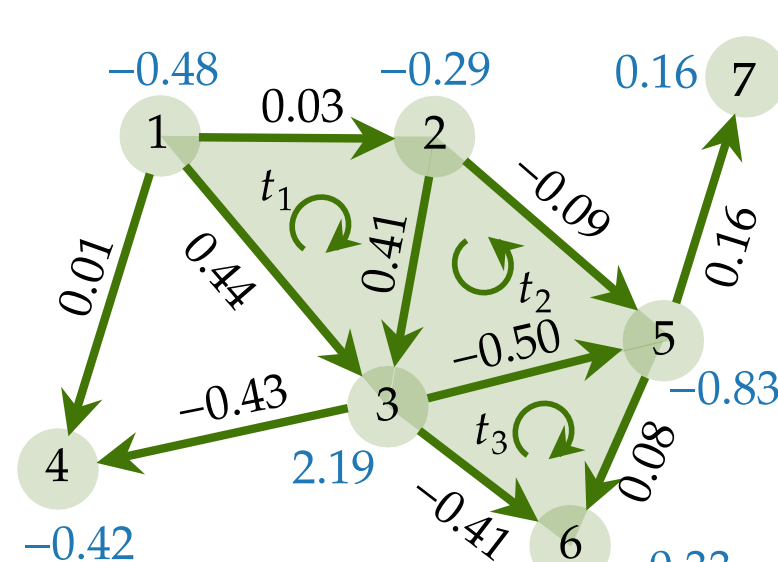
$$\lambda_{G,1} = 0.80$$



$$\lambda_{G,2} = 1.61$$



$$\lambda_{G,5} = 5.12$$

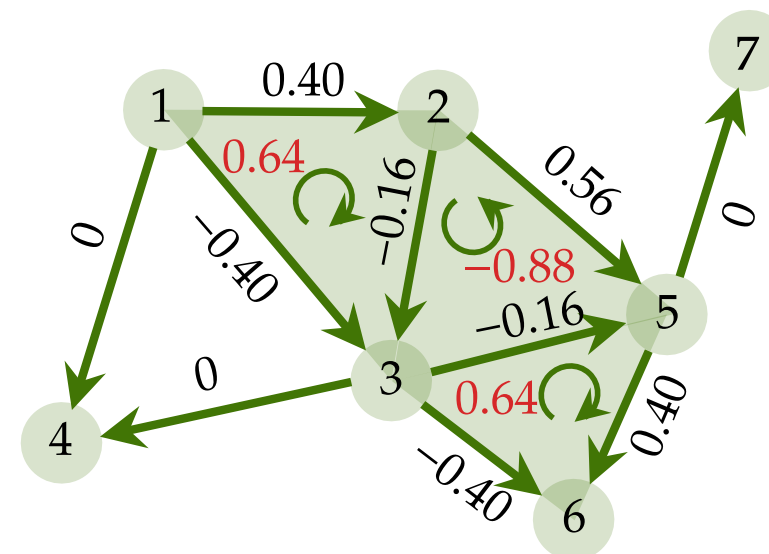


$$\lambda_{G,6} = 6.08$$

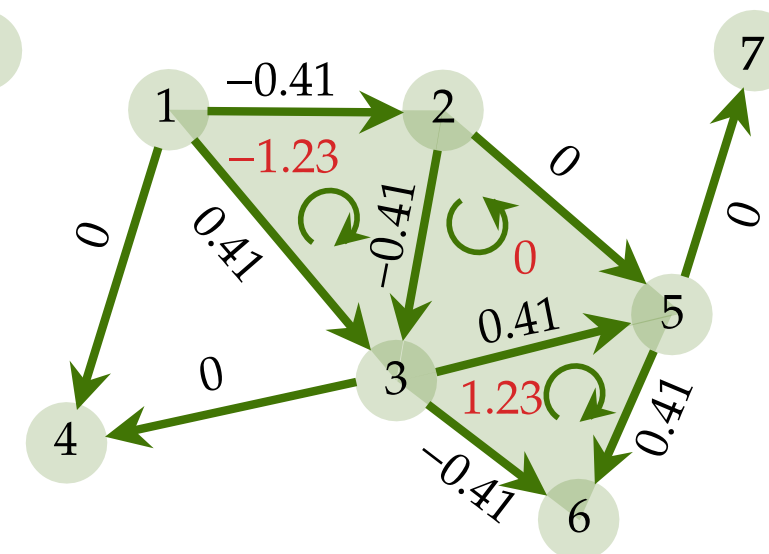
$$\lambda_C = \|\mathbf{B}_2^T \mathbf{u}_C\|_2^2$$

Curl eigenvector

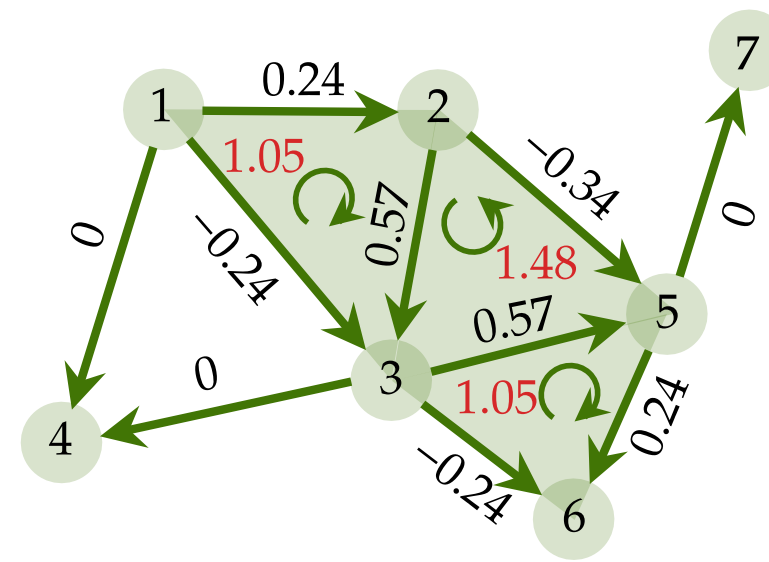
Fourier basis reflecting **rotational** properties



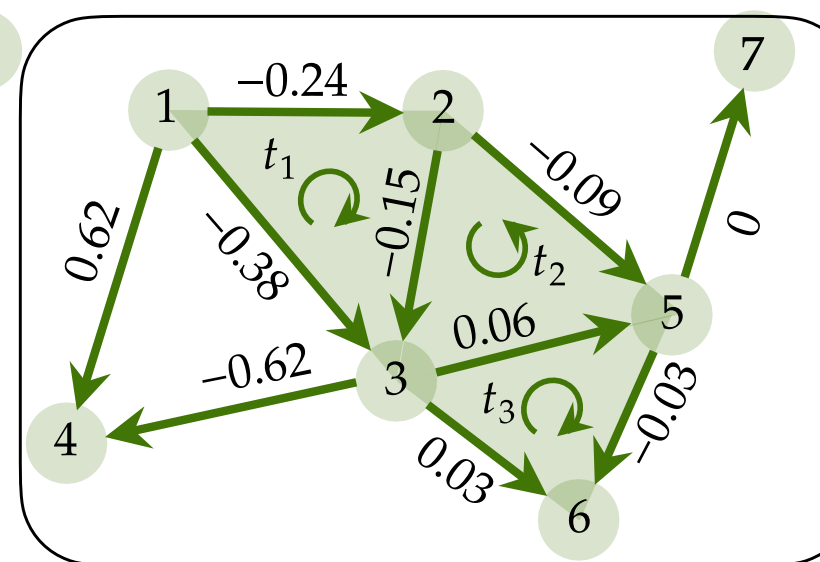
$$\lambda_{C,1} = 1.59$$



$$\lambda_{C,2} = 3.00$$



$$\lambda_{C,3} = 4.41$$



$$\lambda_{H,1} = 0$$

$$\text{EVD: } \mathbf{L}_1 = \mathbf{U}_1 \mathbf{\Lambda}_1 \mathbf{U}_1^T$$

$$\mathbf{U}_1 = [\mathbf{U}_H \ \mathbf{U}_G \ \mathbf{U}_C]$$

$$\text{span}(\mathbf{U}_H) = \ker(\mathbf{L}_1)$$

$$\text{span}(\mathbf{U}_G) = \text{im}(\mathbf{B}_1^T)$$

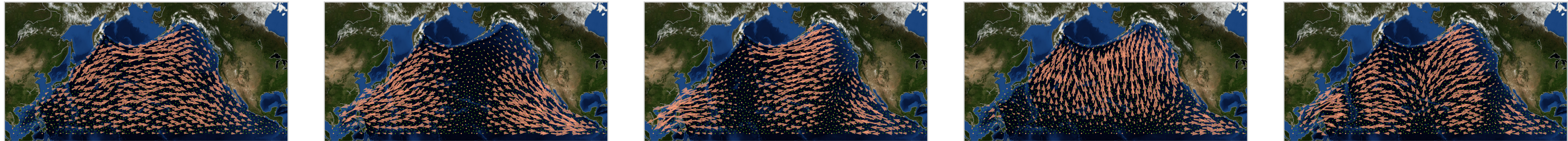
$$\text{span}(\mathbf{U}_C) = \text{im}(\mathbf{B}_2)$$

$$\tilde{\mathbf{x}}_k = \mathbf{U}_k^T \mathbf{x}_k, k = 1$$

$$\tilde{\mathbf{x}}_k = [\tilde{\mathbf{x}}_{k,H}^T, \tilde{\mathbf{x}}_{k,G}^T, \tilde{\mathbf{x}}_{k,C}^T]$$

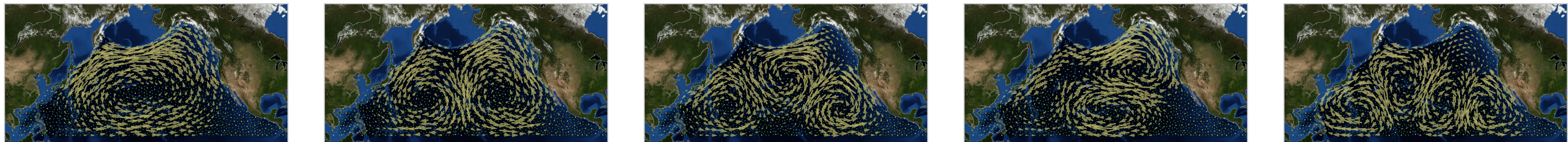
Continuous visualization

- **Down** Laplacian, its nonzero eigenspace spans the **gradient** space



λ_G , more divergent

- **Up** Laplacian, its nonzero eigenspace spans the **curl** space



λ_C , more rotational

Chen, Yu-Chia et al. (2021) "Helmholtzian Eigenmap."